AP PRECALCULUS

Unit 1 • Lesson 1.12 • Guided notes

Topic 1.12 • Transformations of Functions

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Describe translations, dilations and reflections, and their order.
- Write g in terms of f from a description, a table or a graph.
- Find the domain and range of a transformed function and the image of a point.

Vocabulary. translation, vertical dilation, horizontal dilation, reflection, image of a point, domain, range.

1. The four parameters

Key idea

For $g(x) = a\,f\big(b(x-h)\big) + k$:

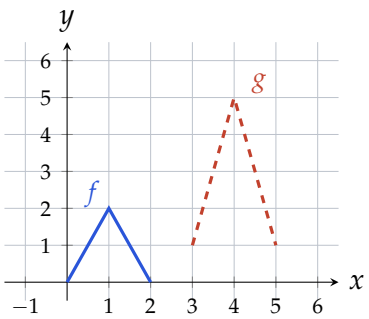
- k: vertical translation, _____ if $k > 0$. h: horizontal translation, _____ if $h > 0$.
- a: vertical dilation by a factor of _____; if $a < 0$, also a reflection over the _____.
- b: horizontal dilation by a factor of _____; if $b < 0$, also a reflection over the _____.

A point (x, y) on f moves to _____ on g. Horizontal changes affect the _____; vertical changes affect the _____.

Example 1 • Describe the transformations

Describe how to obtain $g(x) = -2(x+3)^2 + 5$ from $f(x) = x^2$.

Example 2 • From a graph



Write g in terms of f.

Example 3 • Domain, range and points

f has domain $[-4,2]$ and range $[-1,3]$, and $(2,3)$ is on its graph. For $g(x) = -f(2x)+1$, find the domain, the range and the image of $(2,3)$.

Example 4 • From a table

x	-2	-1	0	1	2
$f(x)$	5	2	1	2	5
$g(x)$	2	1	2	5	10

Write g in terms of f.

Common mistakes

- Shifting the wrong way: $f(x+3)$ moves the graph *left*.
- Thinking $f(2x)$ stretches horizontally; it compresses by a factor of $\frac{1}{2}$.

Check your understanding

1. $(2,5)$ is on f. Then _____ is on $y = f(x+1)-3$.
2. $y = -\frac{1}{2}f(x)$ is a vertical dilation by ____ and a reflection over the _____.